

1. (11 pts) Evaluate the following

$$\int_{\pi/6}^{\pi/4} \int_{\csc \theta}^{\sqrt{3} \sec \theta} \frac{\sec \theta}{r^2} dA + \int_{\pi/4}^{\pi/3} \int_{\sec \theta}^{\sqrt{3} \csc \theta} \frac{\sec \theta}{r^2} dA.$$

2. (11 pts) Compute $\iiint_T xy dV$, where T is the solid tetrahedron with vertices $(0, 0, 0)$, $(1, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 5)$.

3. (11 pts) Evaluate $\int_C (x + y) ds$, where C is the line segment from $(2, 3)$ to $(5, 7)$.

4. (11 pts) Find the average value of the function $f(x, y) = xe^y$ on the unit circle centered at $(0, 0)$.

5. (12 pts) For the vector field $\mathbf{F}$ compute the circulation on the boundary of the region R :

$$\mathbf{F} = \langle x + y^2, x^2 - y \rangle, \text{ where } R = \{(x, y) : 3y^2 \leq x \leq 36 - y^2\}.$$

6. (10 pts) Find the volume of the solid that is outside the cone $\varphi = \frac{\pi}{4}$ and inside the sphere $\rho = 4 \cos \varphi$.

7. (12 pts) Compute the line integral of

$$\mathbf{F}(x, y, z) = \frac{2xy}{z^3} \mathbf{i} + \left(2y - z^2 + \frac{x^2}{z^3}\right) \mathbf{j} - \left(4z^3 + 2yz + \frac{3x^2y}{z^4}\right) \mathbf{k}$$

along the helix given by $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ from $\frac{\pi}{2} \leq t \leq \pi$.

8. (12 pts) Evaluate the following integral

$$\iiint_E z \sqrt{x^2 + y^2} dV,$$

where E is bounded by the plane $z = x + y$ and surface $z = \sqrt{x^2 + y^2}$ within $x^2 + y^2 = 1$ in the first octant.

9. (10 pts) $f(x)$ is a continuous function on $[0, 1]$ and $\int_0^1 f(x) dx = a$, where a is constant. Evaluate the following double integral

$$\int_0^1 \int_x^1 f(x) f(y) dy dx.$$